Commutative Algebra Problem Set #7

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Problem (Exercise 1)**.** *Let A be a valuation ring with fractional field K and residue field L, and denote by $\pi : A \to L$ the quotient map. Let B be the valuation ring with fraction field L and residue field E. Consider the subring $C \subseteq A$ defined by*

$$C = \{x \in A : \pi(x) \in B\}.$$

Prove that C is a valuation ring with fraction field K and residue field E.

Solution. To better visualize the problem, consider the diagram below:

$$\begin{array}{ccccc} K & & & & \\ \uparrow & & & & \\ (A, \mathfrak{m}_A) & \xrightarrow{\pi} & L & & \\ \uparrow & & \uparrow & & \\ C & \xrightarrow{\pi} & (B, \mathfrak{m}_B) & \xrightarrow{\pi'} & E \end{array}$$

Fraction field. First, let us show that $\mathrm{Frac}(C) = K$. Note that the inclusion $\mathrm{Frac}(C) \subseteq K$ is trivial, so we have to show $K \subseteq \mathrm{Frac}(C)$. Informally, this inclusion means that whenever we have $a_0/a_1 \in K$ for $a_0, a_1 \in A$, $a_1 \neq 0$, we can multiply both numerator and denominator by some $s \neq 0$ to get $a_0 s/a_1 s \in \mathrm{Frac}(C)$ (thus $a_0 s, a_1 s \in C$). We formally encapsulate this observation in the following claim.

Claim #1. $\mathrm{Frac}(C) = K$ iff for every $a \in A$ exists $s \in C \setminus \{0\}$ such that $as \in C$.

Let us show this holds for our scenario. Take any $a \in A$. If $\pi(a) \in B$, we automatically win (then, $a \in C$ and thus take $s = 1$). Otherwise, if $\pi(a) \notin B$, $\pi(a)^{-1} \in B$ since B is the valuation ring with $\mathrm{Frac}(B) = L$. Then, $\pi(a)b = 1$ for some $b \in B$. Since map $C \to B$ is surjective, there is some $s \in C$ such that $\pi(s) = b$. Then, $\pi(a)\pi(s) = \pi(as) = 1$, meaning $as - 1 \in \ker(\pi) \subseteq C$. Thus, $as \in C$, as was required to show.

Valuation Ring. Let us now show that C is a valuation ring: that is, for every $x \in K^\times$, either $x \in C$ or $x^{-1} \in C$. Indeed, take any $x \in K^\times$. Since A is a valuation ring, either $x \in A$ or $x^{-1} \in A$. Suppose $x \notin A$, then $x^{-1} \in \mathfrak{m}_A$ (since x^{-1} is non-invertible). Thus, $\pi(x^{-1}) = 0 \in B$ and so $x^{-1} \in C$.

Suppose otherwise that $x \in A$. If $\pi(x) \in B$, we win. Otherwise, $\pi(x)^{-1} \in B$. Thus, $xs \in 1 + \mathfrak{m}_A$ similar to discussion above. Thus, $x \in A^\times$ (if that was not the case, $1 \in \mathfrak{m}_A$ which is a contradiction) and so $x^{-1} \in A$. Thus, $\pi(x^{-1}) = \pi(x)^{-1} \in B$ and so $x^{-1} \in C$.

Maximal Ideal. Finally, denote by $\mathfrak{m}_C$ the maximal ideal of C (since we have proven that C is a valuation ring, C is local). We shall show that $C/\mathfrak{m}_C \cong E$, where the isomorphism is induced from the map $\sigma := \pi' \circ \pi : C \to E$. Indeed, note that both π and π' are surjective, thus so is their composition. From surjectivity follows $\mathrm{im}(\sigma) = E$.

Finally, we want to find kernel. Evidently, $\ker(\sigma) = \pi^{-1}(\mathfrak{m}_B)$. We claim that $\pi^{-1}(\mathfrak{m}_B) \cong \mathfrak{m}_C$. This immediately follows from the following fact.

Claim #2. $c \in C^\times$ iff $\pi(c) \in B^\times$.

Proof of claim. If $cc' = 1$, then $\pi(c)\pi(c') = 1$, so $\pi(c) \in B^\times$. Suppose now that $\pi(c) \in B^\times$, then $\pi(c) \neq 0$ and so $c \notin \ker(\pi) = \mathfrak{m}_A$, thus $c \in A^\times$. In such case $\pi(c)^{-1} = \pi(c^{-1}) \in B$, thus $c^{-1} \in C$, and so c is invertible.

Thus, $C/\ker(\sigma) \cong \mathrm{im}(\sigma)$ which is equivalent to $C/\mathfrak{m}_C \cong E$. □

Problem (Exercise 2). *Let $\zeta = e^{2\pi i/5}$ be the primitive root of unity of order 5 and let $K = \mathbb{Q}(\zeta)$. Prove that the ring of integers $\mathcal{O}_K$ (the integral closure of $\mathbb{Z}$ in K) is equal to $\mathbb{Z}[\zeta]$.*

Solution. We shall show two inclusions. Additionally, nothing restricts us from considering a bit more general case for $\zeta = e^{2\pi i/p}$ where p is prime.

$\mathbb{Z}[\zeta] \subseteq \mathcal{O}_K$. Note that $\zeta \in \mathcal{O}_K$ since it is a root of the pth cyclotomic polynomial $\Phi_p(T) = 1 + T + T^2 + \cdots + T^{p-1}$. Thus, the basis of $\mathbb{Z}[\zeta]$ given by $\{1, \zeta, \ldots, \zeta^{p-2}\}$ also lies in $\mathcal{O}_K$ (since $\mathcal{O}_K$ is a ring). Finally, since any element of $\mathbb{Z}[\zeta]$ is a $\mathbb{Z}$-linear combination of these basis elements, it is also in $\mathcal{O}_K$.

$\mathcal{O}_K \subseteq \mathbb{Z}[\zeta]$. This part is unexpectedly hard. We need to show that if element $\alpha = \sum_{j=0}^{p-2} a_j \zeta^j \in \mathcal{O}_K$ with $a_j \in \mathbb{Q}$, then $a_j \in \mathbb{Z}$. My first attempt was to consider traces of elements of form $\mathrm{Tr}(\zeta^{-i}\alpha)$ (which must be integers), showing $pa_i - \sum_{j=0}^{p-2} a_p \in \mathbb{Z}$. However, this only helped me to deduce $a_j \in \frac{1}{p}\mathbb{Z}$. Then I was stuck.

After searching various discussions on `mathoverflow`, I have found some literature which includes the proof of this fact. Two notable sources are "Algebraic Number Theory" by Jürgen Neukirch (p.58–60, 10.1 and 10.2) and "Algebraic Theory of Numbers" by Pierre Samuel (p.42–44, Theorem 2). Both books notices the special structure of a principal ideal (λ) for $\lambda = 1 - \zeta$. However, it was personally very hard for me to follow Neukirch's version of the proof. A more elementary and straightfoward proof is imo given in Samuel, which I insert below.

The proof consists of showing three small facts, and then combining them together to show that $a_j \in \mathbb{Z}$.

Fact #1. $\mathrm{Tr}(\zeta^j) = p - 1$ if $j = 0$ and -1 otherwise.

This follows from the direct computation of $\mathrm{Tr}(\zeta^j) = \sum_{i=0}^{p-1} \zeta^{ij}$.

Fact #2. $(1-\zeta)\cap\mathbb{Z} = p\mathbb{Z}$ (where $(1-\zeta)$ is the principal ideal in $\mathcal{O}_K$).

Proof. First one notices $p = \prod_{j=1}^{p-1}(1-\zeta^j)$, and in particular $p \in (1-\zeta)$. Thus, $p\mathbb{Z} \subseteq (1-\zeta)\cap\mathbb{Z}$. Assume now that $p\mathbb{Z} \neq (1-\zeta)\cap\mathbb{Z}$. Since $p\mathbb{Z}$ is the maximal ideal in $\mathbb{Z}$, $(1-\zeta)\cap\mathbb{Z} = \mathbb{Z}$, thus $1-\zeta$ is a unit in $\mathcal{O}_K$. But in such case, all conjugates* $1-\zeta^j$ are also units, and thus so is p in $\mathcal{O}_K \cap \mathbb{Z}$. Thus, $p^{-1} \in \mathbb{Z}$. Contradiction.

Fact #3. For any $\beta \in \mathcal{O}_K$, $\mathrm{Tr}(\beta(1-\zeta)) \in p\mathbb{Z}$.

Proof. Note that $\mathrm{Tr}(\beta(1-\zeta))$ is the sum of all conjugates $\beta_j(1-\zeta^j)$. Notice that each such conjugate is a multiple of $1-\zeta^j$ which in turn is a multiple of $1-\zeta$. Thus, $\mathrm{Tr}(\beta(1-\zeta)) \in (1-\zeta)$. But since $\mathrm{Tr}(\cdot)$ of an element of $\mathcal{O}_K$ is an integer, $\mathrm{Tr}(\beta(1-\zeta)) \in \mathbb{Z}\cap(1-\zeta) = p\mathbb{Z}$, where the equality follows from **Fact #2**.

Proof of $\mathcal{O}_K \subseteq \mathbb{Z}[\zeta]$. Take any $\alpha = a_0 + a_1\zeta + \cdots + a_{p-2}\zeta^{p-2} \in \mathcal{O}_K$. Then,

$$\alpha(1-\zeta) = a_0(1-\zeta) + a_1(\zeta-\zeta^2) + \cdots + a_{p-2}(\zeta^{p-2}-\zeta^{p-1}).$$

By applying $\mathrm{Tr}(\cdot)$ to both sides and using **Fact #1**, one obtains $\mathrm{Tr}(\alpha(1-\zeta)) = a_0\mathrm{Tr}(1-\zeta) = pa_0$. But invoking **Fact #3**, one gets $pa_0 \in p\mathbb{Z}$, so $a_0 \in \mathbb{Z}$. Now,

$$(\alpha - a_0)\zeta^{-1} = a_1 + a_2\zeta + \cdots + a_{p-2}\zeta^{p-3}$$

and $(\alpha-a_0)\zeta^{-1}$ is clearly in $\mathcal{O}_K$. Thus, we can apply the same argument to deduce $a_1 \in \mathbb{Z}$. We continue for all a_j. □

Problem (Exercise 3)**.** *Let K be an algebraically closed field and let R be a finitely generated K-algebra. Suppose that for every maximal ideal $\mathfrak{m} \subseteq R$ we have $R_{\mathfrak{m}} = K$. Prove that $R \cong K^n$ for some $n \geq 0$.*

Solution. The solution seems to be quite straightforward after I decided to take a look at Artinian rings... We use the following fact.

Proposition (Stacks Project, Lemma 10.60.5[a])**.** *A Noetherian ring where every prime ideal is maximal is Artinian.*

[a] https://stacks.math.columbia.edu/tag/00KH

Now, recall that prime ideals of $R_{\mathfrak{m}}$ are in one-to-one correspondence with the prime ideals of R contained in $\mathfrak{m}$:

$$\{(0)\} = \mathrm{Spec}(R_{\mathfrak{m}}) \longleftrightarrow \{\mathfrak{p} \in \mathrm{Spec}(R) \mid \mathfrak{p} \subseteq \mathfrak{m}\},$$

thus $\{\mathfrak{p} \in \mathrm{Spec}(R) \mid \mathfrak{p} \subseteq \mathfrak{m}\}$ contains only one element, being $\mathfrak{m}$ itself. Thus, by proposition, R is Artinian. Next, we use the following:

*We recall that the conjugates of α are $\{\sigma(\alpha) : \sigma \in \mathrm{Gal}(K/\mathbb{Q})\}$.

Proposition (Stacks Project, Lemma 10.53.3[a]). *Artinian ring has only finitely many maximal ideals.*

[a] https://stacks.math.columbia.edu/tag/00J4

Thus, suppose $\mathfrak{m}_1, \dots, \mathfrak{m}_n$ are maximal ideals of R. We prove that these ideals are comaximal.

Comaximality. Assume on the contrary $\mathfrak{m}_i + \mathfrak{m}_j \subseteq \mathfrak{n} \subsetneq R$. Localize at $\mathfrak{n}$: $(\mathfrak{m}_i + \mathfrak{m}_j)R_{\mathfrak{n}} \subseteq \mathfrak{n}R_{\mathfrak{n}}$. Since $R_{\mathfrak{n}}$ is a field and $\mathfrak{n}R_{\mathfrak{n}}$ is the maximal ideal of the local ring $R_{\mathfrak{n}}$, we have $\mathfrak{n}R_{\mathfrak{n}} = 0$ and so $(\mathfrak{m}_i + \mathfrak{m}_j)R_{\mathfrak{n}} \subseteq 0$. Thus, both $\mathfrak{m}_i R_{\mathfrak{n}} = 0$ and $\mathfrak{m}_j R_{\mathfrak{n}} = 0$, but then $\mathfrak{m}_i = \mathfrak{m}_j = \mathfrak{n}$. Contradiction.

Thus, by Chinese remainder theorem,

$$R \cong R/\bigcap_{i=1}^{n} \mathfrak{m}_i \cong \prod_{i=1}^{n} R/\mathfrak{m}_i \cong K^n.$$

Here, we used weak Nullstellensatz for $R/\mathfrak{m}_i \cong K$ and the fact that $\bigcap_{i=1}^{n} \mathfrak{m}_i = 0$ (not sure how to prove the latter formally, but roughly the argument is as follows: suppose $x \in \bigcap_{i=1}^{n} \mathfrak{m}_i$, then by localizing at every $\mathfrak{m}_i$ we get $x = 0$ in $R_{\mathfrak{m}_i}$, meaning that $x = 0$ globally). □

Problem (Exercise 4). *Let k be a field and let $R = k[\![T_1, \dots, T_n]\!]$ be the ring of formal power series in $n \geq 1$ variables. Prove that R is a local ring with the maximal ideal $(T_1, \dots, T_n)$.*

Solution. Before considering the general setting, suppose $n = 1$ for a moment. To show that $k[\![T]\!]$ is a local ring, we need to show that all non-units are contained in the ideal (T). Indeed, notice that an element in $k[\![T]\!]$ is invertible iff the constant term is invertible, but for the field, this implies that it is non-zero (more explicitly, the inverse for $f = \sum_{n\geq 0} f_n T^n$ is given by $g = \sum_{n\geq 0} g_n T^n$ where $g_0 = f_0^{-1}$ and $g_n = -f_0^{-1} \sum_{i=1}^{n} f_i g_{n-i}$). Thus, all non-invertible elements must be divisible by T and thus form an ideal (T).

Now we move to the general case by induction. Suppose $k[\![T_1, \dots, T_{n-1}]\!]$ is a local ring with the maximal ideal $(T_1, \dots, T_{n-1})$. Now notice that $k[\![T_1, \dots, T_n]\!] \cong k[\![T_1, \dots, T_{n-1}]\!][\![T_n]\!]$ and thus element in $k[\![T_1, \dots, T_n]\!]$ is non-unit iff the first coefficient (from $k[\![T_1, \dots, T_{n-1}]\!]$) is non-unit, which means that this coefficient lies in $(T_1, \dots, T_{n-1})$. Thus, any non-unit $\varphi \in k[\![T_1, \dots, T_n]\!]$ has a form

$$\varphi = \sum_{k\geq 0} f_k T_n^k = \underbrace{\sum_{i=1}^{n-1} h_i T_i}_{f_0 \in (T_1, \dots, T_{n-1})} + \underbrace{T_n \sum_{k\geq 0} f_{k+1} T_n^k}_{\in (T_n)}, \quad f_i, h_i \in k[\![T_1, \dots, T_{n-1}]\!],$$

which is clearly in $(T_1, \dots, T_n)$. □

Problem (Exercise 5)**.** *Prove that the element $Y^2 - X^2(X+1)$ of the power series ring $\mathbb{C}[\![X, Y]\!]$ is the product of two non-units.*

Solution. Note that in the ring $\mathbb{C}[X, Y]$ the ideal $(Y^2 - X^2(X+1))$ is prime and thus such decomposition is impossible. However, in $\mathbb{C}[\![X, Y]\!]$ we can hope that we can find some two *infinite* series that will cancel out all terms except for X^2, X^3, and Y^2 in the product.

Now, it is very tempting to use the "difference of two squares" formula:

$$Y^2 - X^2(X+1) = (Y - X\sqrt{X+1})(Y + X\sqrt{X+1}).$$

And well, turns out that this is apparently a valid step to do! Notice that:

$$\sqrt{1+X} = 1 + \frac{X}{2} - \frac{X^2}{8} + \frac{X^3}{16} - \frac{5X^4}{128} + \dots,$$

which lies in $\mathbb{C}[\![X]\!]$. Consequently, both $Y - X\sqrt{X+1}$ and $Y + X\sqrt{X+1}$ lie in $\mathbb{C}[\![X, Y]\!]$. Moreover, they are non-units since they lie in (X, Y). □

Problem (Exercise 6)**.** *Let p be a prime and let*

$$A^\circ = \mathbb{Z}_p\langle T\rangle = \varprojlim_n (\mathbb{Z}/p^n\mathbb{Z})[T]$$

be the p-adic completion of $\mathbb{Z}[T]$. Let

$$A = \mathbb{Q}_p\langle T\rangle = A^\circ[1/p].$$

Prove that

$$A \cong \left\{ f = \sum_{n\geq 0} a_n T^n \in \mathbb{Q}_p[\![T]\!] \,:\, |a_n|_p \to 0 \text{ as } n \to \infty \right\}.$$

Solution. By definition, A° is the subring of $\prod_n (\mathbb{Z}/p^n\mathbb{Z})[T]$ such that for each $(f_1, f_2, f_3, \dots)$ one has $f_n \equiv f_{n+1} \pmod{p^n}$. This condition deciphers as follows: if $f_n = \sum_{i\geq 0} f_{n,i} T^i$, $f_{n,i} \in \mathbb{Z}/p^n\mathbb{Z}$ and $f_{n+1} = \sum_{i\geq 0} f_{n+1,i} T^i$, $f_{n+1,i} \in \mathbb{Z}/p^{n+1}\mathbb{Z}$, then one has $f_{n,i} \equiv f_{n+1,i} \pmod{p^n}$. This way, the sequence $(f_{1,i}, f_{2,i}, \dots)$ represents some p-adic integer $f_i \in \mathbb{Z}_p$ by definition. This way, we can associate any element $f \in A^\circ$ with the power series $\sum_{i\geq 0} a_i T^i$ for $a_i = (f_{n,i})_{n\geq 0} \in \mathbb{Z}_p$. Here is the neat visualization below:

	T^0	T^1	T^2	T^3	…	
$f_1 \in (\mathbb{Z}/p)[T]$	$f_{1,0}$	$f_{1,1}$	0	0	…	Degree d_1
$f_2 \in (\mathbb{Z}/p^2)[T]$	$f_{2,0}$	$f_{2,1}$	$f_{2,2}$	0	…	Degree d_2
$f_3 \in (\mathbb{Z}/p^3)[T]$	$f_{3,0}$	$f_{3,1}$	$f_{3,2}$	$f_{3,3}$	…	Degree d_3
⋮	↓	↓	↓	↓	⋱	↓
Limit $f \in \mathbb{Z}_p\langle T\rangle$	a_0	a_1	a_2	a_3	…	$\lvert a_i\rvert_p \to 0$

However, of course not any element of $\mathbb{Z}_p[\![T]\!]$ belongs to A°: we need to find some condition for coefficients of this infinite series. For now, we have not used that each f_n has a finite number of terms: that is, $f_{n,i} = a_{i,n} \equiv 0 \pmod{p^n}$ starting from some $i \geq d_n$ for each $n > 0$. By means of staring at the figure above, one notices $v_p(a_i) \geq n$ for all $i > d_n$, implying $|a_i|_p \leq p^{-n}$ for all $i > d_n$, implying that $|a_i|_p \to 0$ for $i \to \infty$. Thus,

$$\mathbb{Z}_p\langle T\rangle \cong \left\{ \sum_{n\geq 0} a_n T^n \in \mathbb{Z}_p[\![T]\!] \,:\, |a_n|_p \to 0 \text{ as } n \to \infty \right\}.$$

Now we localize: $A = A^\circ[1/p]$. In principle, this should not drastically change the representation given above. Indeed, note that any element $g \in A$ can be written as $g = p^{-m} f$ for some $f \in A^\circ$ and some integer $m \geq 0$. Using representation above, let $f = \sum_{n\geq 0} a_n T^n$ with $\lim_{n\to\infty} |a_n|_p = 0$. Then, $g = \sum_{n\geq 0}(p^{-m}a_n)T^n$ with now $p^{-m}a_n \in \mathbb{Q}_p$. But note that p^{-m} is just some constant, so we still have that $\lim_{n\to\infty} |p^{-m}a_n|_p = 0$.

What remains to show is the reverse inclusion: if it so happens that for some $g = \sum_{n\geq 0} a_n T^n \in \mathbb{Q}_p[\![T]\!]$ we have $\lim_{n\to\infty} |a_n|_p = 0$, then $g \in A$. But notice that the limit being equal to 0 means the sequence is bounded: that is, $|a_n|_p \leq p^m$ for all n and some fixed large enough m. Then, $|p^m a_n|_p \leq 1$ and thus $p^m a_n \in \mathbb{Z}_p$ for all n. Thus, $g = p^{-m} f$ where $f = \sum_{n\geq 0}(p^m a_n)T^n \in A^\circ$ as $\lim_{n\to\infty} |p^m a_n| = 0$. □